Juana Granados Rodríguez

Curriculum Vitae et Studiorum

Personal Information

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Nationality: Spanish, Colombian

Education and Training

Universidad de los Andes — Bogotá, Colombia

Bachelor's Degree in Physics *(GPA: 4.28/5.00)* — 2020 - 2024

- Relevant coursework: Electromagnetism, Quantum Mechanics, Renormalization, Statistical Physics, Computational Methods, Particle Physics.
- Recipient of the *Préstamo Condonable para Carreras Especiales* by the Universidad de los Andes and fully condoned due to outstanding academic performance.

Minor in Mathematics

- Relevant coursework: Discrete Mathematics, Advanced Linear Algebra, Abstract Algebra, Complex Variable Calculus, Probability.

Projects and Research Experience

Nanomagnetism Group and Seminar — Universidad de los Andes

Active Participation in Seminar — September 2023 - December 2024

- Presented results during seminar and group meetings.

Experimental Project — September 2023 - December 2023

- Developed a project investigating the magnetic and electric properties of Co-doped V_2O_5 nanoparticles under the supervision of Professor Juan Gabriel Ramirez Rojas, Ph.D., Associate Professor at Universidad de los Andes.
- Conducted experiments using Vibrating Sample Magnetometry (VSM), X-ray Diffraction (XRD), and Raman Spectroscopy for material characterization.
- Analyzed experimental data and contributed to interpreting of results in the context of nanomagnetism.

Research Assistant — June 2024 - Present

- Collaborating on research focused on studying the magnetic and electric properties of Co-doped V_2O_5 nanoparticles.
- Conducted experiments such as VSM, XRD, and Raman Spectroscopy for material characterization; analyzed experimental data and contributed to interpreting results.
- Contributing to the preparation and submission of a scientific paper for publication in a peer-reviewed journal.

Quantum Field Theory and Mathematical Physics Seminar — Universidad de los Andes

Active Participation in Seminar — September 2023 - July 2024

- Contributed to discussions on advanced topics in quantum field theory.

- Shared advancements in my learning and collaborated with peers and professors to deepen my understanding of quantum field theory.

Undergraduate Thesis January 2024 - August 2024

- Analyzed deeply the mathematical structure of wavefront sets in quantum field theory and applied Epstein-Glaser techniques to explore renormalization processes in theoretical physics under the supervision of Professor Andres Fernando Reyes Lega, Ph.D, Associate Professor at Universidad de los Andes. View Document

Work Experience

Teaching Assistant - Honors Integral Calculus and Differential Equations

Universidad de los Andes, Bogotá, Colombia August 2021 - December 2021

- Graded assignments and exams, assisted the professor in virtual classes and developing class materials and evaluations, addressed students' questions online, and managed the course page and resources.

Teaching Assistant - Vector Calculus

Universidad de los Andes, Bogotá, Colombia June 2023 - August 2023

- Graded assignments and exams, held regular office hours to guide students through challenging concepts, and provided academic support to ensure their understanding and progress.

Skills and Interests

Programming and Technical Tools: Python (NumPy, SciPy, Matplotlib, Pandas, Scikit-Learn), LaTeX, R, Git, GitHub, Jupyter Notebooks

Data Analysis and Tools: Computational modeling, data visualization, statistical analysis, Python for data analysis, experience with VSM, XRD, and Raman spectroscopy

Experimental Skills: Familiar with laboratory protocols, vibrating Sample Magnetometry (VSM), X-ray Diffraction (XRD), Raman Spectroscopy, and nanoparticle synthesis techniques.

Languages: Spanish (Native), English (IELTS Academic 7.5 band score)

Personal Attributes: Communication of complex scientific concepts, strong analytical skills, effective teamwork in academic research settings, adaptability to new technologies for data analysis and research in physics.